

Obesity in Adolescents

TO THE EDITOR: Regarding the Clinical Practice article by Hannon and Arslanian (July 20 issue)¹: adolescent obesity, with its substantial socioeconomic effects, demands specialized solutions.² Innovative therapies such as glucagon-like peptide 1 agonists are emerging, although their long-term effects are yet to be fully known.³ As members of the Bariatric Endoscopy Committee of the International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO), we highlight the potential of endoscopic procedures, notably endoscopic sleeve gastropasty, for this group. The Multicenter ESG Randomized Interventional Trial (MERIT) indicates that endoscopic sleeve gastropasty is a low-risk intervention that is superior to lifestyle changes and repeatable if needed.⁴ Unlike gastric bypass or sleeve gastrectomy, endoscopic sleeve gastropasty is not associated with long-term metabolic complications such as nutritional deficiencies, “dumping syndrome,” or reflux risks.⁵ Given that adolescents have a longer life expectancy than adults, safe and effective interventions such as endoscopic sleeve gastropasty are crucial, while allowing for any bariatric procedures that may be warranted in the future. Early research on endoscopic intervention in this cohort is promising, and ongoing studies may further establish their usefulness. We advocate for increased awareness of these interventions and for their integration into the treatment algorithm for adolescent obesity.⁶

Barham Abu Dayyeh, M.D., Ph.D.

Mayo Clinic
Rochester, MN

Christine Stier, M.D., Dr.Med.

Mannheim University
Mannheim, Germany

for the IFSO Bariatric Endoscopy Committee

Dr. Abu Dayyeh reports receiving consulting fees and research support from Apollo Endosurgery and Boston Scientific, research support from USGI Medical, Spatz Medical, and EndoGastric Solutions, and consulting fees from Olympus and Medtronic. Dr. Stier reports receiving consulting fees from Johnson & Johnson, Novo Nordisk, Nitinotes, Caranx Medical, and Apollo Endosurgery, being a board member of the German Obesity Alliance (a nonprofit organization), and being chair of the Bariatric Endoscopy Committee of the International Federation for the Surgery of Obesity and Metabolic Disorder (a

nonprofit organization). No other potential conflict of interest relevant to this letter was reported.

1. Hannon TS, Arslanian SA. Obesity in adolescents. *N Engl J Med* 2023;389:251-61.
2. Ling J, Chen S, Zahry NR, Kao T-SA. Economic burden of childhood overweight and obesity: a systematic review and meta-analysis. *Obes Rev* 2023;24(2):e13535.
3. Butler PC, Dry S, Elashoff R. GLP-1-based therapy for diabetes: what you do not know can hurt you. *Diabetes Care* 2010;33:453-5.
4. Abu Dayyeh BK, Bazerbachi F, Vargas EJ, et al. Endoscopic sleeve gastropasty for treatment of class 1 and 2 obesity (MERIT): a prospective, multicentre, randomised trial. *Lancet* 2022;400:441-51.
5. Bhandari M, Kosta S, Reddy M, Mathur W, Neto MG, Bhandari M. Four-year outcomes for endoscopic sleeve gastropasty from a single centre in India. *J Minim Access Surg* 2023;39:101-6.
6. Alqahtani A, Elahmedi M, Alqahtani YA, Al-Darwish A. Endoscopic sleeve gastropasty in 109 consecutive children and adolescents with obesity: two-year outcomes of a new modality. *Am J Gastroenterol* 2019;114:1857-62.

DOI: 10.1056/NEJMc2309715

TO THE EDITOR: The article by Hannon and Arslanian is timely and appropriately discusses the health consequences of obesity in adolescents. However, the renal perspective in this context is missing. Visceral adipocytes secrete adipocytokines with deleterious effects, including endothelial dysfunction, oxidative stress, and activation of the renin-angiotensin system. These factors result in increased renal blood flow, glomerular hyperfiltration, subnephrotic proteinuria, and subsequent development of focal segmental glomerulosclerosis.^{1,2} Obese persons have large kidneys and glomerulomegaly. Obesity-related glomerulopathy is a distinct entity that leads to progressive glomerulosclerosis and a decline in renal function.³ In the Framingham Offspring Study, higher body-mass index was associated with increased risk of chronic kidney disease during a mean follow-up of 18.5 years.⁴ It is important to perform urine examination for albuminuria as well as estimation of the glomerular filtration rate (GFR) at baseline and periodically thereafter in adolescents with obesity to guide appropriate treatment strategies.

Subham Bhowmik, M.B., B.S.

All India Institute of Medical Sciences
New Delhi, India
subhamdb@gmail.com

Himansu S. Mahapatra, M.D.

Dr. Ram Manohar Lohia Hospital
New Delhi, India

No potential conflict of interest relevant to this letter was reported.

1. Bhowmik D, Tiwari SC. Metabolic syndrome and chronic kidney disease. *Indian J Nephrol* 2008;18:1-4.
2. Tang SCW, Sharma K. Pathogenesis, clinical manifestations, and natural history of diabetic kidney disease. In: Johnson RJ, Floege J, Tonelli M, eds. *Comprehensive clinical nephrology*. 7th ed. Philadelphia: Elsevier, 2024:363-79.
3. D'Agati VD, Chagnac A, de Vries AP, et al. Obesity-related glomerulopathy: clinical and pathologic characteristics and pathogenesis. *Nat Rev Nephrol* 2016;12:453-71.
4. Fox CS, Larson MG, Leip EP, Culleton B, Wilson PWF, Levy D. Predictors of new-onset kidney disease in a community-based population. *JAMA* 2004;291:844-50.

DOI: 10.1056/NEJMc2309715

THE AUTHORS REPLY: Abu Dayyeh and Stier, on behalf of the IFSO Bariatric Endoscopy Committee, advocate for endoscopic sleeve gastropasty as a low-risk intervention for the treatment of severe (class 2 or 3) adolescent obesity. This procedure uses an endoscopic approach to place full-thickness sutures that restrict stomach volume, is reversible, and is repeatable if weight loss is not maintained. MERIT, which showed greater weight loss with endoscopic sleeve gastropasty plus lifestyle modifications than with lifestyle modifications alone (total body-weight loss at 1 year, 13.6% vs. 0.8%), did not include adolescents but rather involved persons 21 to 65 years of age.¹ Data for adolescents remain very limited. In one case series, investigators performed endoscopic sleeve gastropasty in 109 patients of Middle Eastern descent 10 to 21 years of age (91.7% female), some of whom had obesity-associated coexisting conditions. The mean ($\pm$ SD) percentage total weight loss at 6, 12, and 24 months was 14.4 $\pm$ 6.5%, 16.2 $\pm$ 8.3%, and 13.7 $\pm$ 8.0%, respectively; however, only 43 and 17 participants were evaluated at 1 and 2 years, respectively.² There were no severe or unanticipated adverse events. Although such preliminary data may appear promising, more data are needed in pediatric populations (particularly from randomized, controlled trials) before any strong recommendations could be made regarding the use of endoscopic sleeve gastropasty in the management of pediatric obesity.

Bhowmik and Mahapatra point out important renal implications of adolescent obesity, which are often overlooked. Indeed, we regret that we did not include them in our article. Nephropathy is often attributed to hyperglycemia in persons with both obesity and diabetes; however, obesity is an independent risk factor for glomerulopathy and the development of chronic kidney disease.³ We agree that it is important to perform a urine examination for albuminuria and to estimate the GFR at baseline and periodically thereafter to identify the presence of or progression to kidney disease. Caloric restriction with high-protein diets is not recommended, because high protein consumption overall is associated with glomerular hyperfiltration.⁴ Obesity-associated hyperfiltration has been shown to improve with treatments that result in weight loss.⁵

Tamara S. Hannon, M.D.

Indiana University School of Medicine
Indianapolis, IN

Silva A. Arslanian, M.D.

UPMC Children's Hospital of Pittsburgh
Pittsburgh, PA
silva.arslanian@chp.edu

Since publication of their article, the authors report no further potential conflict of interest.

1. Abu Dayyeh BK, Bazerbachi F, Vargas EJ, et al. Endoscopic sleeve gastropasty for treatment of class 1 and 2 obesity (MERIT): a prospective, multicentre, randomised trial. *Lancet* 2022;400:441-51.
2. Alqahtani A, Elahmedi M, Alqahtani YA, Al-Darwish A. Endoscopic sleeve gastropasty in 109 consecutive children and adolescents with obesity: two-year outcomes of a new modality. *Am J Gastroenterol* 2019;114:1857-62.
3. Adebayo OC, Nkoy AB, van den Heuvel LP, et al. Glomerular hyperfiltration. 2. Clinical significance in children. *Pediatr Nephrol* 2023;38:2529-47.
4. Kamper A-L, Strandgaard S. Long-term effects of high protein diets on renal function. *Annu Rev Nutr* 2017;37:347-69.
5. Nehus EJ, Khoury JC, Inge TH, et al. Kidney outcomes three years after bariatric surgery in severely obese adolescents. *Kidney Int* 2017;91:451-8.

DOI: 10.1056/NEJMc2309715

Correspondence Copyright © 2023 Massachusetts Medical Society.

THE JOURNAL'S WEB ADDRESSES

To submit a letter to the Editor or to obtain information about the status of a submitted manuscript:
authors.NEJM.org

For general information on the Journal's editorial policies:
nejm.org/about-nejm/editorial-policies

The Journal's website: NEJM.org